Presentation Plan: Does Air Quality Explain Adult Asthma?

Overview

- **Audience:** TruBridge Health Analytics reviewers and healthcare stakeholders who want to know whether air quality should guide asthma prevention. Mixed technical background.
- **Goal:** Communicate one clear finding and what to do next, in 7 slides.
- **Core message:** Air quality does not explain adult asthma across U.S. counties. County size and state matter more, so decisions need neighborhood-level data.
- **Approach:** Insight-first. The answer appears on slide 1. There is no slide per analysis: the statistical tests are merged into one finding.

Storyline

1. Air quality does not explain adult asthma across U.S. counties (title and answer)
2. We tested whether counties with cleaner air report less adult asthma, using 769 U.S. counties
3. Counties with worse air do not report more asthma, and every test agrees
4. County size and state matter more than air quality
5. Los Angeles has high AQI but below-average asthma, so county averages can mislead
6. These results show the limits of county data, not that air pollution is harmless
7. Neighborhood-level data, income measures, and child asthma are the next steps

Slide Details

Slide 1: Title and answer

- **Action title:** Air quality does not explain adult asthma across U.S. counties
- **Body:**
    - Subtitle: a county-level analysis of CDC and EPA data
    - One-line answer: counties with worse air do not report more asthma
    - Presenter, externship, and date
- **Visual:** clean title slide with one large number, r = -0.04 (the correlation between AQI and asthma)
- **Speaker notes:** Lead with the answer so the audience knows where the talk is going.

Slide 2: Question and data

- **Action title:** We tested whether counties with cleaner air report less adult asthma, using 769 U.S. counties
- **Body:**
    - Why: asthma affects millions of adults, air pollution is a known trigger, and the Los

Angeles area has long faced air-quality challenges
    - Data: CDC PLACES (modeled adult asthma, 2025) joined to EPA median AQI (2026)
    - Method: correlation, group comparisons, and regression with state effects
- **Visual:** simple flow diagram of two data sources, then merge, then 769 counties, then analysis
- **Speaker notes:** Keep this to about one minute. The evidence is on the next slides.

Slide 3: Main finding

- **Action title:** Counties with worse air do not report more asthma, and every test agrees
- **Body:**
    - Correlation of -0.04 between median AQI and asthma prevalence
    - Lowest vs. highest AQI quartile: 10.67% vs. 10.61%
    - t-test, Mann-Whitney, ANOVA, chi-square, and regression all find no difference (p-values from 0.19 to 0.82)
- **Visual:** Figure 2 (scatter plot with a flat trend line), with Los Angeles County highlighted
- **Speaker notes:** The point is the flat line. Figure 3 and Table 1 hold the test details as backup.

Slide 4: County size and state

- **Action title:** County size and state matter more than air quality
- **Body:**
    - Larger counties report slightly less asthma ($r = -0.26$)
    - Adding state raises explained variation from 9% to 67%, while AQI stays non-significant
    - The highest county averages are in Maine, Oregon, West Virginia, Massachusetts, and Oklahoma
- **Visual:** Figure 4 (asthma vs. county population, log scale)
- **Speaker notes:** Part of the state effect comes from how PLACES models small areas, so avoid over-interpreting it.

Slide 5: The Los Angeles area

- **Action title:** Los Angeles has high AQI but below-average asthma, so county averages can mislead
- **Body:**
    - Los Angeles County: AQI 62 and asthma 9.1%, compared with a 10.6% average
    - Orange (9.0%) and San Diego (8.9%) look similar, while Trinity and Humboldt have the highest asthma in California (11.0%)
    - Los Angeles research shows asthma clusters by neighborhood, which a county average hides
- **Visual:** Figure 5 (California counties with Los Angeles highlighted)
- **Speaker notes:** This is the local takeaway: the county number does not describe any one neighborhood.

Slide 6: Limitations

- **Action title:** These results show the limits of county data, not that air pollution is harmless
- **Body:**
  - County averages hide neighborhood exposure, and PLACES covers adults only
  - Only monitored counties (827) are included, which skews toward larger counties
  - Income, race, and smoking are missing, and AQI is not long-term exposure
- **Visual:** none, or three simple icons
- **Speaker notes:** Say plainly that this is a limit of the data, not evidence of safety.

Slide 7: Recommendations and next steps

- **Action title:** Neighborhood-level data, income measures, and child asthma are the next steps
- **Body:**
  - Repeat the analysis with census-tract data in Los Angeles County
  - Add income, poverty, and race and ethnicity from the American Community Survey
  - Study childhood asthma with multi-year PM2.5 data
  - Do not target asthma programs by county AQI alone
- **Visual:** four-step roadmap graphic
- **Speaker notes:** End on the recommendation, then invite questions and suggestions for neighborhood-level air quality sources.

Backup Slides (Not Counted)

- **B1, Distributions:** Figure 1. Asthma averages 10.6% and median AQI is 41.
- **B2, Test details:** Figure 3 (box plots by AQI quartile) and Table 1 (summary of tests).
- **B3, Data cleaning:** 27 unmatched AQI records, 34 suppressed asthma estimates, and 3 counties matched twice.
- **B4, Largest counties:** In the top quarter of counties by population (n = 193), the correlation is r = 0.05 (p = 0.54) and AQI is not significant in regression (p = 0.075). Among the 42 counties with 1 million or more residents, r = 0.19 (p = 0.24). Neither is statistically significant, but both lean positive, which supports a finer-grained follow-up.

Design Notes

- One action title and one visual per slide, with at most four supporting points.
- Reuse the report's figures with a consistent palette: blue for data, orange for Los Angeles.
- Put "Source: CDC PLACES (2025), EPA AirData (2026), author analysis" in the footer of each data slide.
- Slides 1 and 3 carry the answer, so no separate executive summary slide is needed.